UNIT 2: SHAPES AND REGIONS

SESSION 7: HANDLING OCCLUSION

SLO 1: UNDERSTAND AND EXPLAIN THE CHALLENGES POSED BY OCCLUSION IN OBJECT DETECTION AND SHAPE ANALYSIS.

Scenario Analysis

Situation

In a traffic monitoring system, one car partially blocks another car.

Answer the following:

1. **What problem does this create for the object detection system?**

Ans- It creates difficulty in detecting and recognizing the partially hidden object because only part of the object is visible.

2. **Why might the system incorrectly identify the hidden car?**

The system may incorrectly identify the hidden car because important features and shape information are blocked by the other car.

Section 4 – Identify the Challenge

Match the following:

Problem	Description
Partial Occlusion	A. Object completely hidden
Full Occlusion	B. Only part of the object is visible
Detection Error	C. System misidentifies object due to missing information

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Section 5 – Application Thinking

Write one example where occlusion occurs in the following applications.

Application	Example of Occlusion
Traffic monitoring	One vehicle partially blocks another vehicle.
Face recognition	A person's face is partially covered by a mask or another person.

Medical imaging	**One organ or tissue may partially hide another structure in an image.**

Reflection

Explain in one or two sentences:

1. **Reflection – Reflection can create misleading or duplicate visual information in an image, making it difficult for the system to distinguish the actual object from its reflected appearance.**

Why is occlusion considered a major challenge in computer vision?

Occlusion is a major challenge in computer vision because it hides some or all of an object's important features. This makes object detection, recognition, tracking, and shape analysis more difficult and can lead to incorrect identification.